



UNIT 19

ELECTROMAGNETIC INDUCTION



Induced Electromotive force EMF

Induced electromotive force (emf) can be generated in two primary ways: by relative movement and by changing a magnetic field.

(i) By Relative Movement (The Generator Effect):

This method is based on Faraday's Law of Electromagnetic Induction, which states that an emf is induced in a conductor when it experiences a change in magnetic flux. The generator effect, also known as motional emf, occurs when there is a relative motion between a conductor and a magnetic field.

Example:

In a simple electrical generator, a coil of wire rotates within a magnetic field. This induced emf drives an electric current through an external circuit.

(ii) By Changing a Magnetic Field (The Transformer Effect):

This method also relies on Faraday's Law but involves changing the strength or orientation of a magnetic field around a stationary conductor.

Example:

In a transformer, an alternating current in the primary coil creates a changing magnetic field. This is the basis for how transformers transfer electrical energy between circuits at different voltages.

Generator Effect:

Induced emf due to relative motion between a conductor and a magnetic field.

Transformer Effect:

Induced emf due to a changing magnetic field around a stationary conductor.

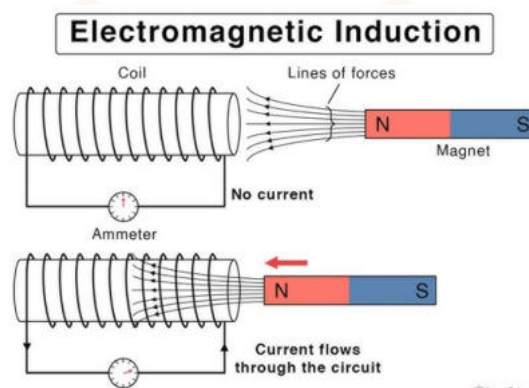
Both effects are fundamental principles in electromagnetism and are widely utilized in various electrical devices and systems.

Faraday's law of electromagnetic induction:

It can be stated as follows:

The electromotive force (emf) induced in a closed circuit is directly proportional to the rate of change of magnetic flux passing through the circuit.

Mathematically, Faraday's law is often expressed as: |



$$\varepsilon = -N \frac{\Delta\phi}{\Delta t} \dots\dots(19. 1)$$

Where ε represents the induced electromotive force (emf) measured in volts (V).

$\frac{\Delta\phi}{\Delta t}$ At Represents the rate of change of magnetic flux with respect to time, it is measured in weber per second (Wb/s V), and N indicates number of turns of the coil: Negative sign is introduced because the induced emf opposes the change in flux. It will be explained in Lenz's law.

Factors affecting the magnitude of the induced emf:

1. Magnetic Flux Change
2. Number of Turns in the Coil
3. Area of the Coil
4. Angle between Magnetic Field and Coil
5. External Factors

Lenz's law and principle of conservation of energy:

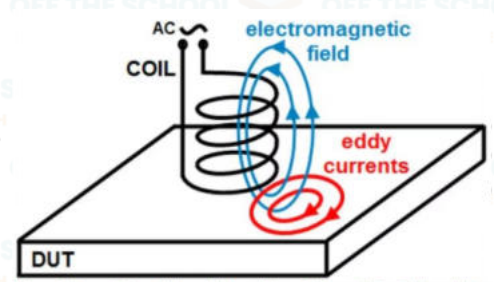
After the introduction of Faraday's law of electromagnetic induction, Heinrich Friedrich Lenz formulated a rule for determining the direction of an induced current within a loop.

According to Lenz's law

"The direction of induced emf in a circuit is always such that it opposes the cause which produces it"

Lenz's law and Conservation of Energy:

When we apply this principle to electromagnetic induction, we see that the work done in changing the magnetic field is converted into electrical energy the form of the induced current. Hence Lenz's law is in accordance with the law of energy, conservation of energy.



Eddy currents and their magnetic and heating effects:

Eddy currents are circulating currents induced in a conductor when it is exposed to a changing magnetic field. They are a common phenomenon in electromagnetic systems. These currents can have both magnetic and heating effects. Now let's discuss about the production of eddy currents and their effects:

Production of Eddy Currents:

Eddy currents are produced according to Faraday's law of electromagnetic induction, which states that a changing magnetic field induces an electromotive force (emf) in a conductor. When a conductor, such as a metal plate or a coil of wire, is subjected to a changing magnetic field, the magnetic flux through the conductor changes, leading to the induction of eddy currents. As shown in figure.

Magnetic Effects of Eddy Currents:

Counteracting Magnetic Field:

Eddy currents generate their own magnetic fields, and the direction of these fields opposes the original magnetic field that induced them. As a result, eddy currents create a magnetic field that counteracts the original magnetic field's change, thereby reducing the net magnetic field.

Magnetic Damping:

In applications like electromagnetic brakes and magnetic dampers, eddy currents are intentionally induced to create a magnetic resistance that opposes motion. This magnetic damping effect is useful for controlling the movement of objects and slowing them down.

Heating Effects of Eddy Currents:

Joule Heating:

Eddy currents experience resistance as they flow through the conductor, and this resistance results in the conversion of electrical energy into heat, following Joule's law. Eddy currents can heat up the conductor. In some cases, such as induction heating for metal processing or cookware, this heating effect is used for practical applications.

Reduction of Eddy Currents:

In many electrical systems, eddy currents represent an undesirable source of energy loss, especially in transformers and electric motors. To minimize these losses, laminated cores are used in transformers to break up the conducting paths and reduce the formation of eddy currents.

Enhanced Efficiency:

By reducing the losses associated with eddy currents, laminated cores improve the overall efficiency of electric motors, generators, and transformers. Less energy is wasted as heat, allowing these devices to operate more efficiently and with less energy consumption.

Mitigation of Vibration and Noise:

Eddy currents generated in a solid iron core can lead to vibrations and noise, which can be undesirable in many applications. Laminated cores help reduce these vibrations and noise levels, making the devices quieter and more mechanically stable.

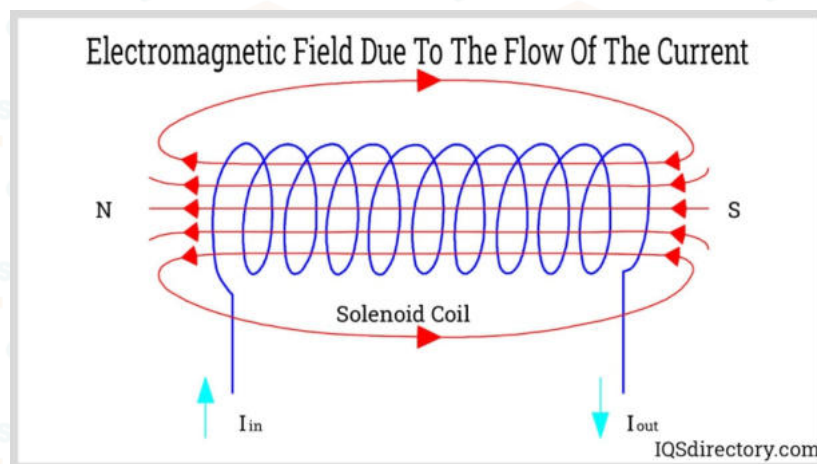
Better Cooling and Thermal Management:

Since laminated cores result in reduced heat generation due to decreased eddy current losses, they often allow for more efficient cooling and thermal management in these devices.

This can lead to longer operational lifetimes and improved reliability.

Self-Induction:

When an electric current passes through a coil, it creates a magnetic field around it. If the current in the coil changes, either increasing or decreasing, the magnetic flux within the coil also changes accordingly. As a result of the change in magnetic flux, an induced emf is generated in the same coil. This process, where a changing current in the coil induces an emf in the coil itself, is known as self-induction.



Consider the circuit shown in Figure, which comprises a coil connected in series with a battery and a rheostat. When the rheostat is adjusted, it changes the current flowing through the circuit, leading to changes in the magnetic flux within the coil.

These variations in magnetic flux result in the induction of an electromotive force (emf) within the coil.

If we denote the magnetic flux through a single loop of the coil as Φ , then the total magnetic flux passing through the N turns of the coil would be $N\Phi$. This relationship holds true because the magnetic flux is directly proportional to the magnetic field, which, in turn, is proportionate to the current (I) flowing through the circuit.

$$N\Phi \propto I$$

$$N\Phi = LI \dots \dots \dots (19.2)$$

Where $L = \frac{N\Phi}{I}$ is constant of proportionality called the self-inductance of coil. It depends on factors such as the coil's number of turns, its cross-sectional area, and the material used for the core. Hence inductance of coil made up of soft iron is greater than the air core coil.

By the Faradays law of electromagnetic induction

$$\varepsilon = -N \frac{\Delta\Phi}{\Delta t} \dots \dots \dots (19.3)$$

Substitute equation (1) in Equation (2), we get

$$\varepsilon = -L \frac{\Delta I}{\Delta t} \dots \dots \dots (19.4)$$

Above equation (19.4) shows that the self-induced electromotive force (emf) within a coil is directly proportional to the rate of change of current within the coil.

Unit of Inductance:

The unit of self-inductance is the Henry (H), named after the American physicist Joseph Henry. One Henry (1 H) is defined as the amount of self-inductance in a circuit when a change in current of 1 ampere per second (1 A/s) induces an electromotive force (emf) of 1 volt (1 V) within the same circuit. Mathematically, this relationship can be expressed as:

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$$1 \text{ Henry} = \frac{1 \text{ Volt}}{1 \text{ Ampere per Second}}$$

Energy Stored in an Inductor:

An inductor is a passive electrical component that stores energy in the form of a magnetic field when an electric current flow through it.

The ability of an inductor to store electric potential energy is based on the fundamental principle of electromagnetic induction.

Here's a simple explanation of how an inductor stores electric potential energy:

Current Flow:

When an electric current flows through a coil of wire (the inductor), it creates a magnetic field around the coil.

Magnetic Field Buildup:

As the current increases, the strength of the magnetic field around the coil also increases. This process takes a short amount of time, as the magnetic field does not build up instantly; it grows with the rate of change of the current.

Energy Storage:

The energy is stored in the magnetic field. The inductor stores electric potential energy in this magnetic field. The more current that flows through the inductor or the faster the current changes, the stronger the magnetic field, and thus, the more energy is stored.

Magnetic Field Collapse:

When the current decreases or stops (like when you turn off the power), the magnetic field starts to collapse. As it collapses, it induces an electromotive force (EMF) or voltage in the coil.

Released Energy:

This induced voltage represents the stored energy being released. The inductor converts the stored magnetic energy back into electric energy. This energy can be used to sustain the current for a short period or can be transferred to other parts of the circuit.

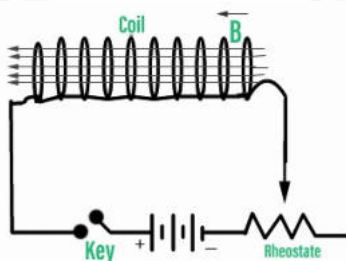
Energy produced in Self Induction:

An inductor stores energy in its magnetic field when it carries a direct current (DC).

This energy remains stored as long as the inductor continues to carry the current.

When the

conversely, it decreases when the current is reduced.



Consider an inductor connected to a DC power source through a switch, as depicted in Figure. When the switch is closed, the current in the inductor gradually rises until it reaches its maximum value, denoted as I . This changing current leads to a corresponding change

in the magnetic flux within the coil, causing an induced electromotive force (emf) to establish itself in the coil. Consequently, an induced current is generated in the circuit, which works to minimize the current produced by the battery in accordance with Lenz's law.

As a result, the battery must perform work on the charges to build up the current I . This work is mathematically expressed as follows:

$$W = \Delta V \Delta q \dots\dots\dots (19.5)$$

As the induced emf is produced in the inductor so $\Delta V = \Delta \Phi$ and using work energy relation above equation can be written as:+

$$E = \frac{\Delta q}{\Delta t} L \Delta I$$

On solving we get

Since,

$$E = \frac{1}{2} L I^2$$

Where:

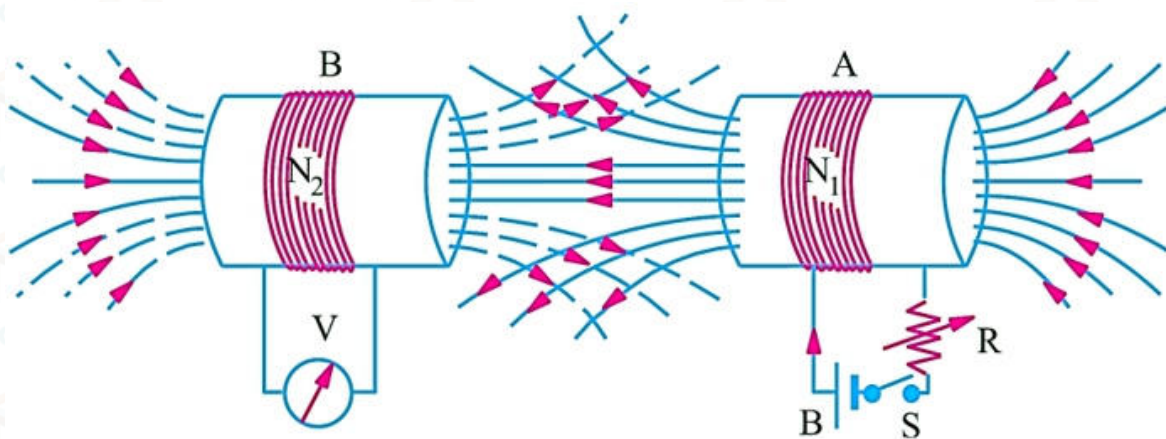
E is the energy stored in joules (J).

L is the self-inductance of the inductor in Henries (H).

I is the current flowing through the inductor in amperes (A).

Mutual Inductance

"A phenomenon where a changing current in one coil induces an electromotive force (EMF) in another coil is known as mutual induction".



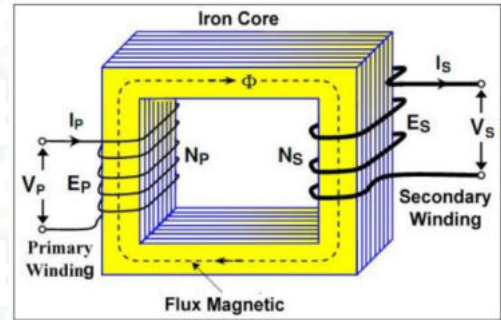
"A phenomenon where a changing current in one coil induces an electromotive force (EMF) in another coil is known as mutual induction"

Transformer:

A transformer is an electrical device used to transfer electrical energy between two or more coils of wire through electromagnetic induction.

Construction of a Transformer:

The transformer consists of following main components as shown in figure



Core:

Transformers consist of two coils of wire, known as the primary coil and the secondary coil, wound around a common magnetic core.

The core is typically made of materials with high magnetic permeability, such as laminated iron or ferrite, to enhance the efficiency of the transformer.

Primary Coil:

Transformer Construction The primary coil is connected to the input voltage V_p source. It consists of a specific number of turns of wire wound around one section of the core denoted as N_p .

Secondary Coil:

The secondary coil is connected to the load or the device that needs the transformed voltage. It has a different number of turns of wire wound around another section of the core denoted by N_s .

Insulation:

Both the primary and secondary coils are insulated from each other to prevent electrical contact and ensure electrical isolation.

Working of a Transformer:

Imagine we have an alternating electromotive force (emf) applied to the primary coil. If, at a certain moment (t), the magnetic flux within the primary coil is changing at a rate of $\frac{\Delta\Phi}{\Delta t}$, this change in flux will induce a counteracting electromotive force (emf) in the primary coil, opposing the applied voltage. We can express the instantaneous value of this self-induced emf as follows:

$$\text{Self-induced emf} = -N_p \frac{\Delta\Phi}{\Delta t}$$

If the coil's resistance can be considered negligible, then the opposing emf in the primary coil is equal in magnitude but opposite in direction to the applied voltage. This relationship can be represented as:

$$V_p = -N_p \frac{\Delta \phi}{\Delta t} \dots \dots \dots (19.10)$$

Here, N_p represents the number of turns in the primary coil.

Now, let's assume that the flux passing through the primary coil also flows through the secondary coil. In other words, these two coils are magnetically linked. Hence, the rate of change of magnetic flux in the secondary coil will also be and the magnitude of the induced emf across the secondary coil can be expressed as:

$$V_s = -N_s \frac{\Delta \phi}{\Delta t} \dots \dots \dots (19.11)$$

Where N_s is number of turns in secondary

When we divide equation (1) by equation (2), we obtain

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Step-Up or Step-Down Transformers:

The ratio of the number of turns in the primary coil (N_p) to the number of turns in the secondary coil (N_s) determines whether the transformer is a step-up or step-down transformer.

If $N_s > N_p$ then it's a step-up transformer and it increases the voltage.

Conversely, if $N_p > N_s$ it is a step-down transformer and it decreases the voltage.

Step up transformer:

Step-up transformers are designed to increase the voltage of electricity.

They have more turns in the secondary coil than in the primary coil, resulting in a higher secondary voltage compared to the primary voltage.

- > Reduced Current
- > Efficient Long-Distance Transmission

Step-Down Transformers:

- > Voltage Reduction
- > Balanced Current

Transmission of Electricity:

Step-down and step-up transformers are used for the electric supply from power station to houses and electric appliances. Here's how they are used in the electric supply process as shown in figure



1. Power Generation at the Station:

Electricity is generated at power stations, often using generators powered by various sources

2. Step-Up Transformers at the Power Station:

To raise the voltage to levels suitable for long-distance transmission, step-up transformers are employed at the power station.

3. Transmission and Substations:

The high-voltage electricity is transmitted over transmission lines to substations located at various points in the electricity distribution network.

4. Step-Down Transformers in Substations:

Step-down transformers reduce the high-voltage electricity to lower, safer voltage levels, suitable for local distribution. This lower voltage is then distributed to houses through distribution lines.

Motional emf:

Motional electromotive force (emf) is a phenomenon that arises when a conductor moves through a magnetic field, inducing an electromotive force within the conductor.

This concept is based on Faraday's law of electromagnetic induction and is a fundamental aspect of electromagnetism.

AC Generator

An AC generator, also known as an alternator, is a device that converts mechanical energy into electrical energy in the form of alternating current (AC). It's based on the principle of electromagnetic induction, which was first described by Michael Faraday.

Construction of an AC Generator:

An AC generator typically consists of the following four components as shown in figure

- I. Armature
- II. Slip-rings
- III. Field magnet
- IV. Brushes

Armature:

The armature or rotor is a rectangular coil mounted on a rotating shaft. It is the brushes component that spins within a magnetic field to induce electrical current.

Permanent magnet:

The permanent magnet or stator is a stationary part of the generator that produces a magnetic field. It is typically made up of a set of magnets or electromagnets arranged in a circular or cylindrical configuration around the rotor.

Slip Rings and Brushes:

The ends of the rotor coil are connected to slip rings, which are conductive rings that rotate with the rotor. Brushes, typically made of graphite, press against the slip rings to collect the generated electrical current.

Shaft and Bearings:

The rotor is mounted on a shaft that allows it to rotate freely. Bearings are used to reduce friction and enable smooth rotation.

Working of an AC Generator:

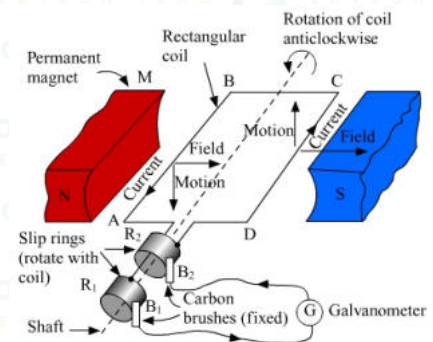
The working of an AC generator involves several steps:

Rotation of the Armature coil:

The coil or rotor is mechanically rotated using an external source of power, such as an engine, a turbine, or any other energy source.

Generation of a Changing Magnetic Field:

As the rotor spins within the stator's magnetic field, the magnetic field within the rotor coil changes. This changing magnetic field induces an electromotive force (emf) or voltage in the coil,



Generation of Alternating Current:

The induced voltage causes an electric current to flow through the coil, and since the magnetic field's direction is changing, the current produced is alternating in nature. This means the direction of the current constantly switches back and forth, resulting in an AC output.

Collection of Output:

The alternating current generated in the rotor coil is collected using slip rings and brushes. The brushes maintain contact with the slip rings as they rotate, allowing the generated AC to be drawn from the generator.

Induced EMF of an AC generator:

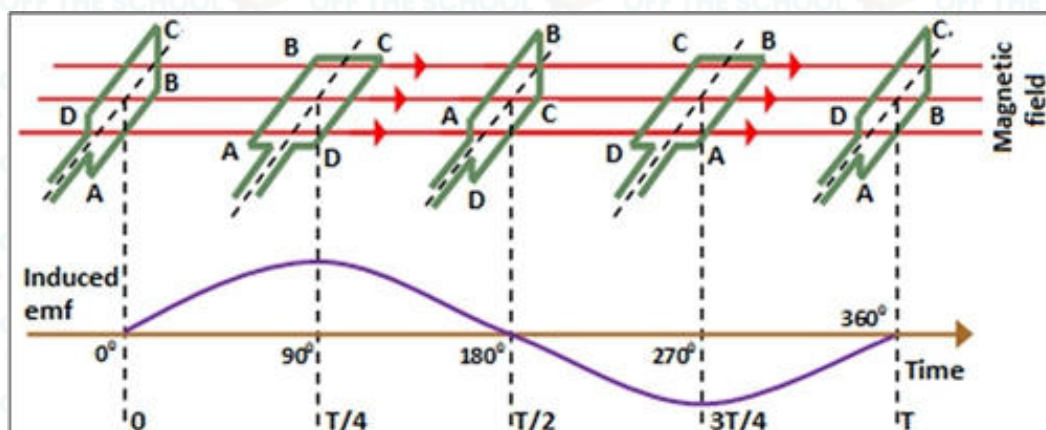
Suppose the armature coil AHCD rotates counterclockwise. As it rotates, the magnetic flux linked with it changes, inducing a current in the coil as shown in figure (a,b). The direction of the induced current follows Fleming's right-hand rule.

When the armature is in the vertical position and rotates counterclockwise, wire AH moves downward, while DC moves upward. This causes the induced electromotive force (emf) to flow from H to A and from D to C, within the coil, forming a current path of DCHA.

In the external circuit, the current flows along BRB2. This current direction remains consistent during the first half-turn of the armature.

However, during the second half-revolution, wire AH moves upward, and wires CD move downward, leading to a reversal in the direction of induced current within the coil to AHCD. In the external circuit, the direction changes to B2RB:. Therefore, the direction of the induced emf and the current alternates after every half revolution in the external circuit as well. Consequently, the current produced switches direction in each cycle.

When the coil is rotated, a motional electromotive force (emf) is generated in each of its sides, but these emfs have opposite directions because the two sides are moving in opposite directions relative to the magnetic field. However, the other two sides of the coil are moving in the same direction with respect to the field.



Consequently, emfs are induced in these sides in the same direction, leading to a mutual cancellation effect. As a result, the total emf induced in the coil is given by

Where

v = linear velocity,

B = magnetic field,

N = Number of turns,

L = length of coil

and ω = angular velocity.

Since each particle of the sides AH and CD rotates in a circle of radius equal to the width of the coil b .

$$v = b/2 \omega$$

$$\varepsilon = 2 b/2 \omega B \sin \omega t$$

$$\varepsilon = ANB \omega \sin \omega t$$

Where $\varepsilon_0 = ANB \omega$ is the maximum or peak value of the emf which depends on the area and number of the coil, intensity of the field and speed of rotation.

In terms of the frequency " f " i.e number of turns per second.

$$\varepsilon = \varepsilon_0 \sin \omega t$$

A.C Motor:

An AC (Alternating Current) motor is a device designed to convert electrical energy into mechanical energy by using alternating current. There are various types of AC motors, but the main features and components are generally consistent across different designs. The primary components as shown in figure and their roles in an AC motor are as follow:

Stator:

The stator is the stationary part of the motor and contains the primary windings. Its main role is to produce a rotating magnetic field when AC voltage is applied. The rotating magnetic field interacts with the rotor to induce motion.

Rotor:

The rotor is the rotating part of the motor. It can be of different types, such as squirrel-cage or wound rotor, depending on the motor design. When the stator produces a rotating magnetic field, the rotor experiences a torque due to the interaction with the field. This torque causes the rotor to rotate and generate mechanical output.

Bearings:

Bearings are essential components that support and allow the rotor to rotate within the stator. They reduce friction, enabling smooth and efficient operation of the motor.

Shaft:

The shaft is connected to the rotor and extends beyond the motor housing. It transfers mechanical energy to the outside world, allowing the motor to perform useful work when connected to a load.

Cooling System:

Many AC motors incorporate cooling systems, such as fans or fins, to dissipate heat generated during operation. Efficient cooling is important to prevent overheating and prolong the motor's lifespan.

The main features and components work together to enable the AC motor to function. When AC voltage is applied to the stator windings, a rotating magnetic field is created, which exerts a torque on the rotor. The rotor's rotation results in mechanical work being performed by the motor, and the motor can be used to drive various mechanical devices, such as fans, pumps, conveyor belts, and more.

The efficiency and performance of an AC motor depend on the design and quality of its components, the load it drives, and its operating conditions. Proper maintenance and control mechanisms, such as speed control, are often used to optimize the motor's performance in different applications.